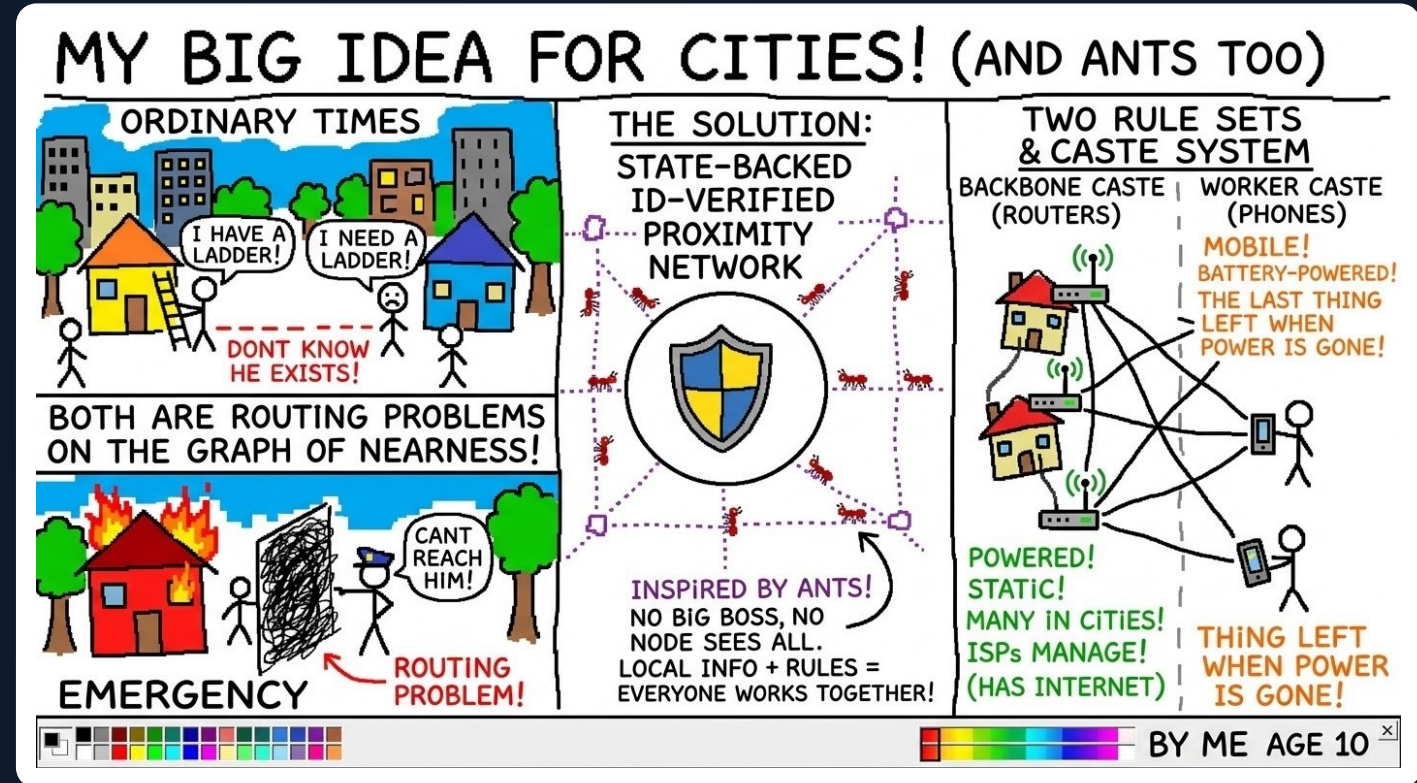


Pomóc

The network that is already there

A dual-use civic mesh: neighbourhood sharing on an ordinary Tuesday, resilient communication on the day the network goes dark.



Team: Apexlab

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Two failures, one root cause



On an ordinary day

Resources sit idle. Nobody nearby knows they exist.

~13 min

a drill is used in its whole lifetime



On the bad day

Help sits unreachable. The mobile network is the first thing to fail.

~80 m

between a cardiac arrest and the AED on the wall

The city already owns what it needs. It is in the wrong pocket.

One network, one identity, two rule sets



Peace mode

- Borrow a drill, get sour cream next door, hand over a parking spot
- Reach is hop-limited (default ~3 hops, configurable): local by protocol, not by policy
- Optional micro-payments or credits



Emergency mode

- Signed official alerts, “I am OK” check-ins, life-critical requests
- No operator network needed: home routers mesh over Wi-Fi, phones relay over Bluetooth LE
- Payments off, priorities on, store-and-forward on

Runs on what the city already has: phones and ISP routers. Routers are the Wi-Fi mesh backbone, phones the Bluetooth edge.

The AED, in ~90 seconds

1

**Ola sends a signed
LIFE_CRITICAL
request with her
location attached.**

It floods the mesh at top priority, pre-empting everything else.

2

**The mesh knows a
registered AED ~40 m
from Marek.**

He is the nearest capable responder and accepts with one tap.

3

**Everyone else sees
the request is taken.**

The mesh guides Marek to the AED, then to Ola.

4

Marek runs it over.

One person moves, the messages did the rest. Staff would take ~6 minutes.

Messages travel hop by hop. The object travels with one person.

Three levels, one signed declaration

	L1 Disruption	L2 Disaster	L3 Security
Cause	cell outage, cable cut, cyberattack	flood, storm, blackout	terror, war, hybrid attack
Entered by	automatic (no backhaul) or local authority	local authority declaration	local authority declaration
Router portal	read + check-in	read + check-in + request	read only
Citizen requests	emergency requests: life-critical, safety, "I am OK"	emergency requests: life-critical, safety, "I am OK"	casualty reporting: voluntary position + short message, sealed to the authorities
Citizen info	on	on	off (rumour control), phones go quiet

Jamming the cells can force L1 at most. Everything above it needs a signature. Portal inputs are aggregated once a minute per router, so the mesh load is bounded by routers, not attackers.

Resource sharing within a few hops

Request → neighbours within the hop limit see it → first accepted match closes it

Shared-ownership amortisation: a tool pays itself back through community use

Registered static resources: public AEDs, shelters, water points pinned to the map

Daily OK (opt-in senior care): no phone use for hours or a detected fall → the phone asks “Are you OK?” → the people the senior chose → only then nearest neighbours and 112

Local questions and answers: is the pharmacy open, who has a spare car seat, is the playground flooded

Guard rails: rate limits per identity, reputation, quiet hours, category mutes

Why it matters for the bad day: the app is already installed, the identities already issued, the graph already known

Example request

Cordless drill, Sunday afternoon

2 PLN

3 hops

expires 2 h

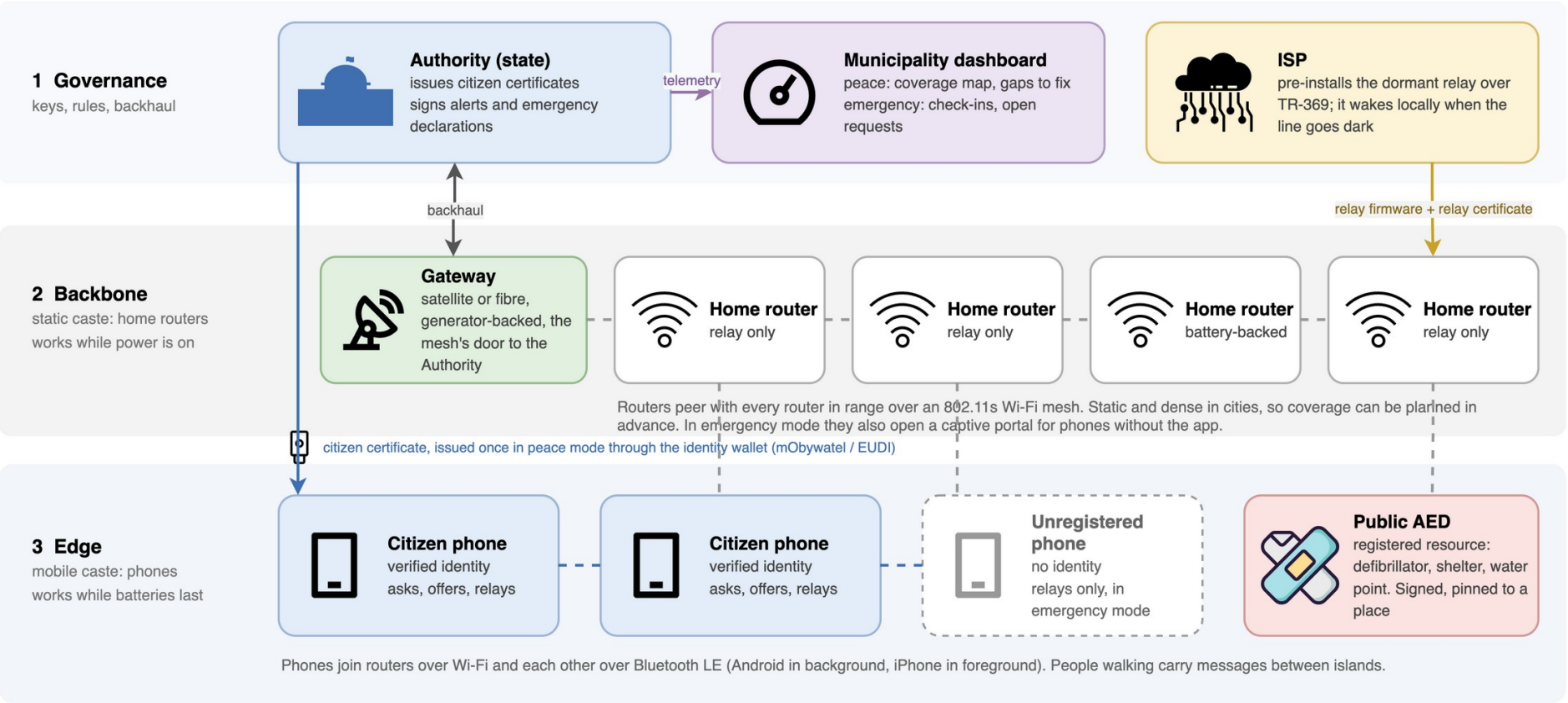
Matched by Marek, **2 hops away**, in ~4 minutes.

No shop trip, no delivery van, one drill working instead of two idle ones.

How the pieces fit together

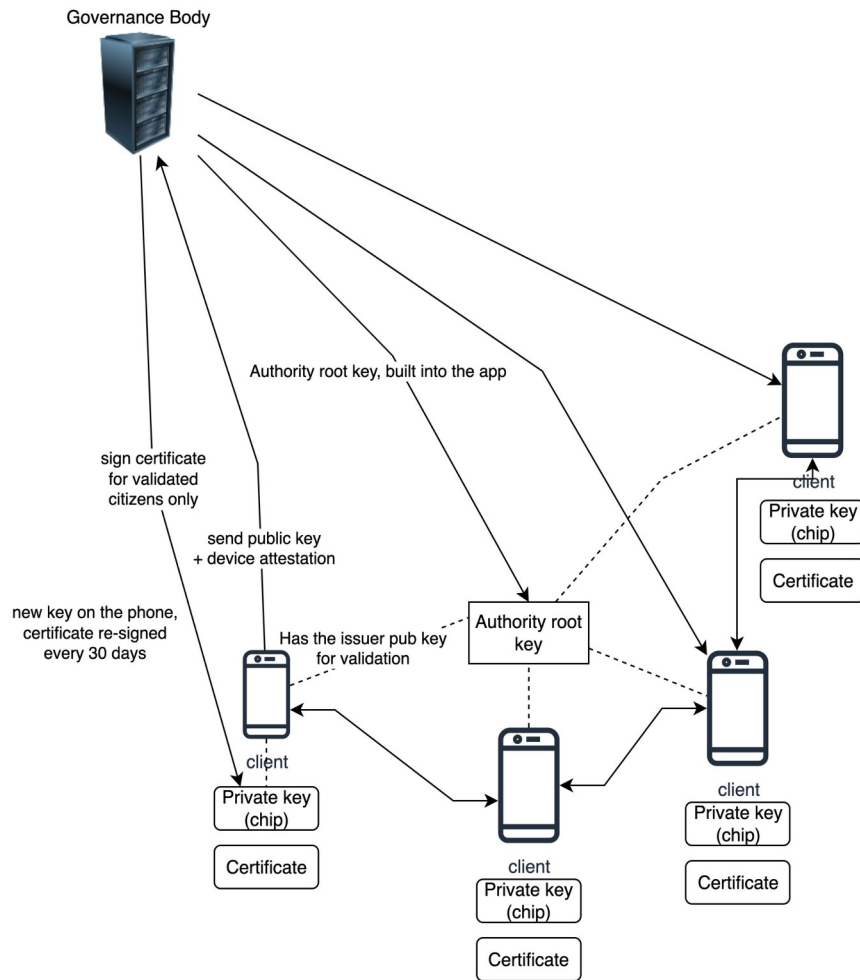
Pomóc

The network that is already there. One protocol, one identity, two rule sets: peace and emergency.



Reading the picture: dashed line = radio neighbours, an edge of the proximity graph · solid arrow = administrative or backhaul relationship · blue = verified citizen · dashed box = no identity, relay only. Only citizens can ask; routers and gateways only carry.

Verified people, verifiable offline



Register once through the national identity wallet (mObywatel / EUDI). The key is born in the phone's secure chip and never leaves it; the Authority verifies the person and the device, then signs a citizen certificate.

Every message is signed. Any node verifies it with the Authority root key already in the app. No network needed.

Routers hold relay certificates: they carry, they can never ask. A cryptographic rule, not a policy.

Pseudonymous on the wire. ~30-day keys: the phone makes a new one and the Authority re-signs it, through the mesh when the internet is gone.

Forged alerts die. Local authorities sign declarations for their area with keys under the Authority root; alerts are signed too. Anything unverifiable is dropped at the first hop.

A simulation of the protocol on a city graph

Kraków district, static router layer plus mobile phone layer, in one interactive one-pager. Deterministic, playable step by step:

1 Peace request + match

2 Flooding: hop limit, TTL, dedup

3 Cells down → L1, router backbone

4 Power down → phone-only, store-and-forward

5 L2 west, L3 east: regional declarations

6 Forged request rejected

Live metrics: reach, delivery per class, hops, latency. Set device counts and radio range and watch the **percolation threshold**: how few nodes are enough.

Honest scope

Built: the protocol simulation (modes, levels, flooding, hop limits, TTL, de-duplication, store-and-forward, degradation)

Not simulated: cryptography is standard and not simulated; the simulation is about mesh behaviour

Not built: router firmware, identity integration, Authority services

Vision, not in the sim: coverage planning. Phone density shifts by district and over time; the city sees where Pomóc is thin and responds with relay nodes, campaigns or incentives

Pomóc is a vision; the simulation is the argument.

Every piece is tried and validated. Nobody has combined them.

00000JAPAN and **Comcast** already flip existing home and public Wi-Fi open in disasters, by remote config

EUDI wallets are mandatory EU-wide: Pomóc plugs into mObywatel in Poland, DÁP in Hungary, any national wallet

Alert RCB is moving into mObywatel; Pomóc is its offline continuation when the cells are down

Japan's Relay-by-Smartphone proved phone-to-phone message relaying after 2011

Peerby: ~80% of neighbourhood requests filled within ~30 minutes

Standard parts: 802.11s router mesh (Freifunk-proven), BLE / Wi-Fi Aware, TR-369, PKI with secure-element keys

The point

The first 24 hours. After an attack or disaster, authorities and the military need time to respond. Pomóc keeps a communication platform alive in that gap.

Fewer casualties. Life-critical requests, casualty reports with voluntarily shared position, and verified official alerts keep moving without cells or power.

Already in every pocket. No new hardware, no new identity scheme: phones, home routers and the national EUDI wallet.

Next: open protocol spec · pilot in one district with one ISP · battery-backed relay placement from the coverage model. Concept, research, docs, simulation and this deck were made with AI assistance (Claude Code; title illustration generated with Gemini); the team stands behind all of it.